



Abstract Art

Getting Abstraction “Just Right”

Jeremy Clark
Developer Betterer
GitHub: [jeremybytes](#)

Level: Intermediate

Visual Studio **LIVE!**
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Data Platform **LIVE!**
INSIGHTS FOR MANAGING YOUR DATA ESTATE

TECHMENTOR
IN-DEPTH TRAINING FOR IT PROS

Artificial
Intelligence **LIVE!**
AI FOR DEVELOPERS AND DATA SCIENTISTS

Cloud &
Containers **LIVE!**
CLOUD-NATIVE, PAAS & SERVERLESS COMPUTING

Cybersecurity
& Ransomware **LIVE!**
DEFENDING AGAINST RANSOMWARE AND OTHER ATTACKS

Session Survey

- Your feedback is very important to us
- Please take a moment to complete the session survey found in the mobile app
- Use the QR code or search for “Converge360 Events” in your app store
- Find this session on the Agenda tab
- Click “Session Evaluation”
- Thank you!





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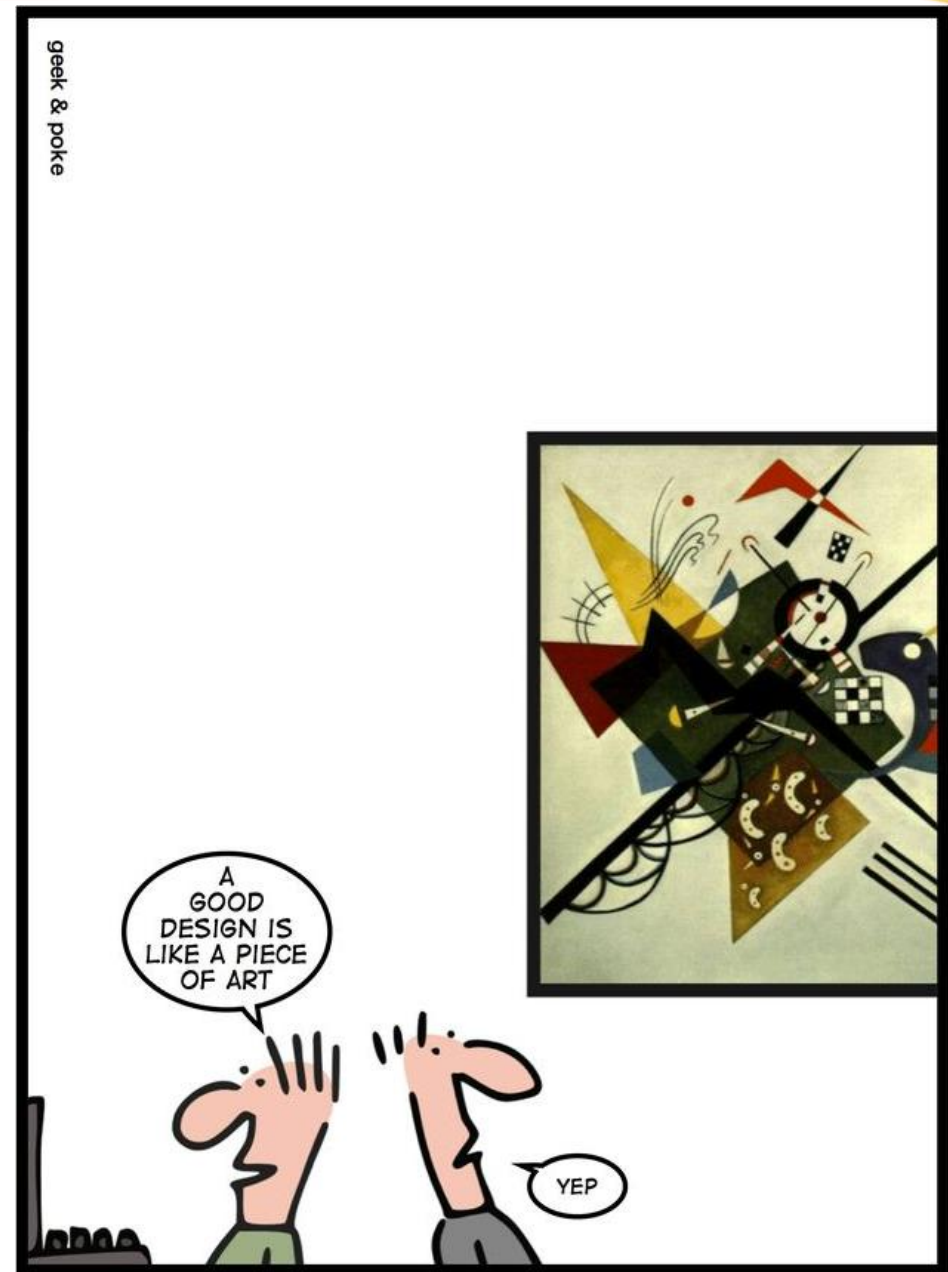
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A Good Design is like A Piece of Art

Geek & Poke – <http://goo.gl/ifu53l>



ABSTRACTION

WHAT IS ABSTRACTION?

Abstraction

- Simplifying reality
- Ignoring extraneous details
- Focusing on what is important for a purpose

ABSTRACTION IS AWESOME!



Maintain



Extend



Test

ABSTRACTION IS AWFUL!



Complexity



Confusion



**Debugging
Difficulty**

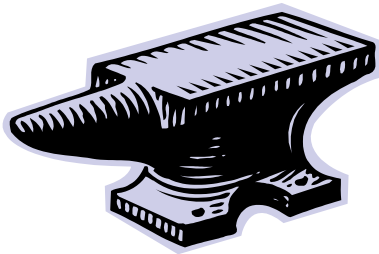
Frustration



https://archive.org/details/goldilocks_and_the_three_bears

GOLDILOCKS THE DEVELOPER

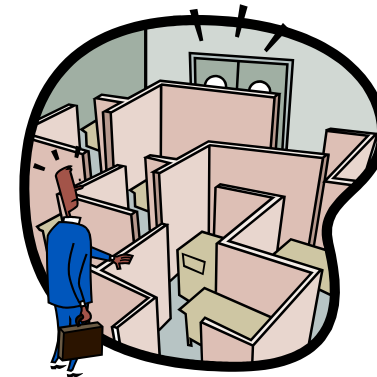
**Too Little
Abstraction**



Just Right



**Too Much
Abstraction**



TWO TYPES OF DEVELOPERS

Over-Abstractor

The diagram consists of two horizontal rows. The top row features a teal rounded rectangle with the text 'Over-Abstractor' in white. To its left and right are thin teal lines that extend to the edges of a larger, empty teal rectangular frame. The bottom row features a blue rounded rectangle with the text 'Under-Abstractor' in white. To its left and right are thin blue lines that extend to the edges of a larger, empty blue rectangular frame.

Under-Abstractor



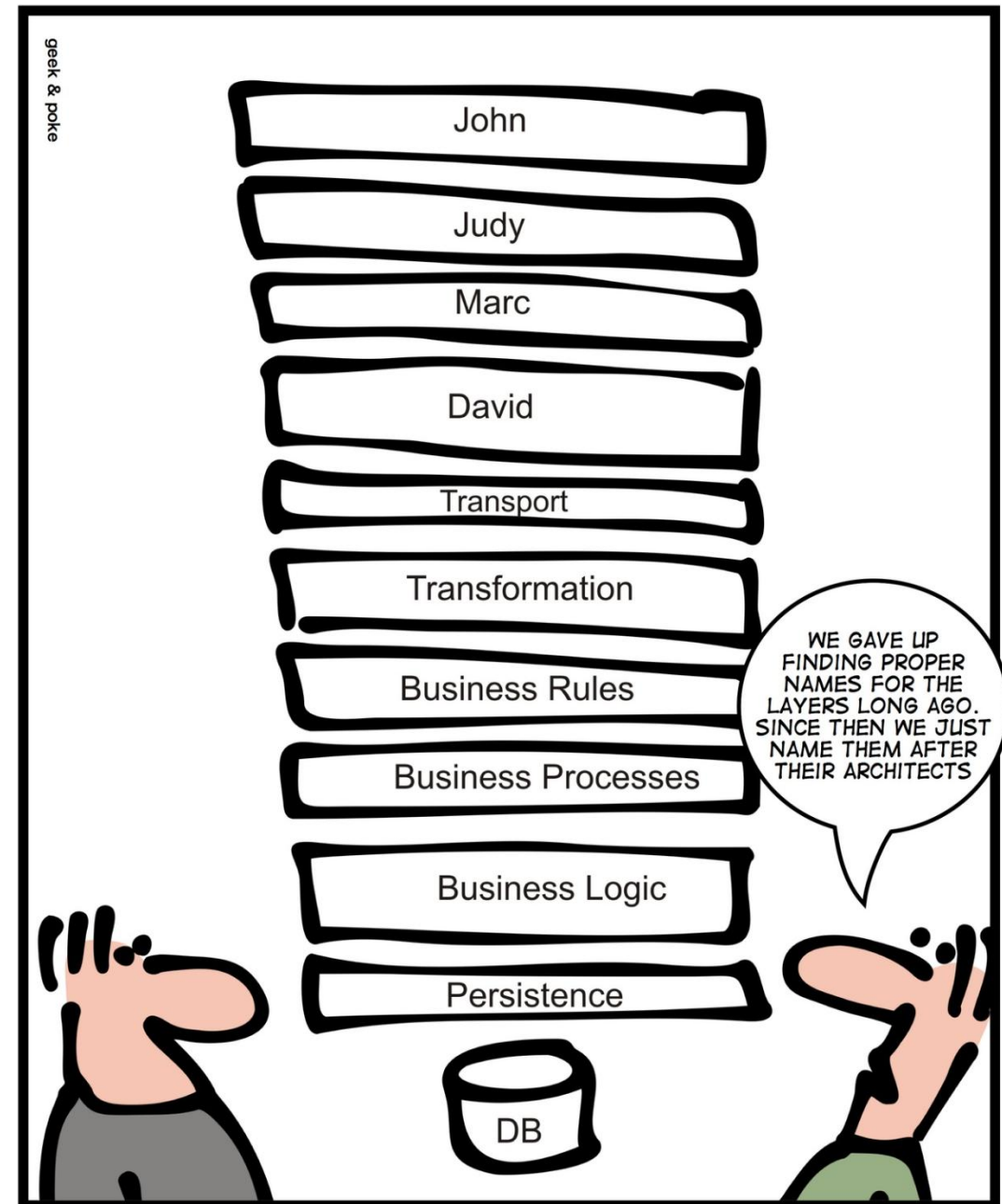
Over-Abstractor

- “We’ll have a good use for this in the future.”
- Overly Complex
- Difficult to Maintain

A Good Architect Leaves A Footprint

Geek & Poke: <http://goo.gl/B4uXa3>

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A GOOD ARCHITECT LEAVES A FOOTPRINT



Under-Abtractor

- “Let’s keep things simple.”
- Rigid
- Difficult to Maintain

COMMON PROBLEM

Over-Abstractor

- “We’ll have a good use for this in the future.”
- Overly Complex
- Difficult to Maintain

Under-Abstractor

- “Let’s keep things simple.”
- Rigid
- Difficult to Maintain



The Default State Quiz

Who Are You?

Let's build a plug-in architecture...

Awesome!
Let's do it.

Maybe we
should look at
compile-time
options.

We need to share a value between modules...

I'll create an object state manager.

Let's use a global variable.

How should we do the UI?

Here's a new
JavaScript
framework.

Let's use the
same framework
we did last time.

Pull data from a database...

ORMs are
awesome!

```
SELECT *  
FROM Customers  
WHERE ID = [@id]
```

We need an object instance...

```
var logger =  
    DIContainer  
    .Resolve<ILogger>()
```

```
var logger =  
    new FileLogger()
```




“
Neither answer is right or wrong. The
correct response is “It depends.”
”

—Jeremy's Standard Response

Let's build a plug-in architecture...

+1

Awesome!
Let's do it.

-1

Maybe we
should look at
compile-time
options.

We need to share a value between modules...

+1

I'll create an object state manager.

-1

Let's use a global variable.

How should we do the UI?

+1

Here's a new
JavaScript
framework.

-1

Let's use the
same framework
we did last time.

Pull data from a database...

+1

ORMs are
awesome!

-1

```
SELECT *  
FROM Customers  
WHERE ID = [@id]
```

We need an object instance...

+1

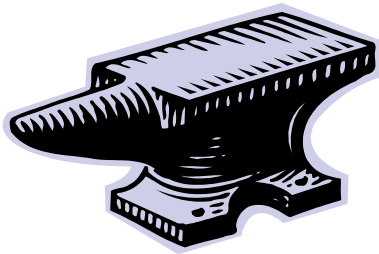
```
var logger =  
    DIContainer  
    .Resolve<ILogger>()
```

-1

```
var logger =  
    new FileLogger()
```

BE HONEST WITH YOURSELF

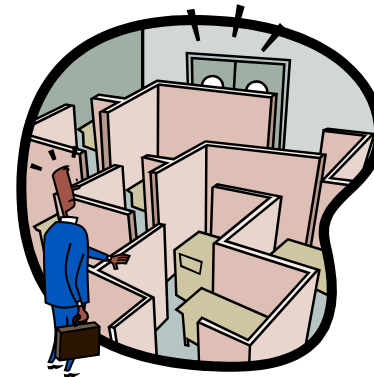
Too Little
Abstraction



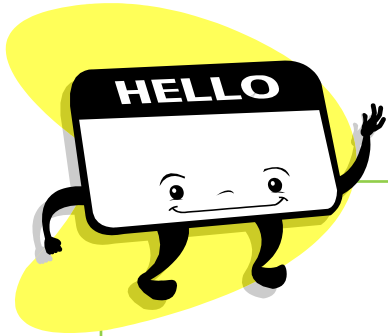
Just Right



Too Much
Abstraction



WHO AM I?



Under-Abstractor

- Hello. My name is Jeremy, and I'm an Under-Abstractor.

“Keep Things Obvious”

“Don't Be Tricky”

REPORTING APPLICATION

The image displays a Reporting Application interface with three overlapping windows. The background window is titled "Report Viewer" and features a sidebar with three buttons: "Report List", "Add Report", and "Update Report", each with a right-pointing arrow icon. The main content area of this window shows a table with two columns: "Report Title" and "Category". The table contains the following data:

Report Title	Category
Quarterly Financials	
Weekly Sales	
Inventory	
Pending Shipments	
Incoming Shipments	
Advertising Sales	

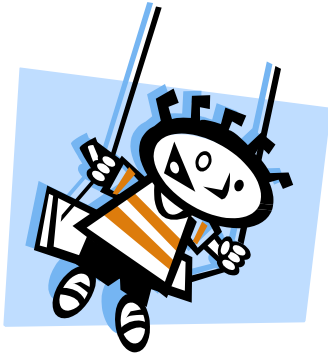
The middle window, also titled "Report Viewer", is titled "Pending Shipments". It contains fields for "Start Date" (8/24/2014), "End Date" (8/24/2014), and "Additional Info". To the right of these fields is a calendar for August 2014.

August 2014						
Su	Mo	Tu	We	Th	Fr	Sa
27	28	29	30	31	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16

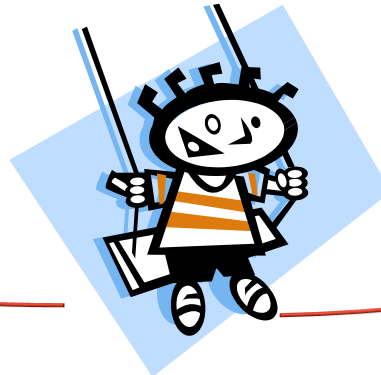
The foreground window is the main application window. It has a menu bar with "New Report File", "Edit Report", and "Edit Parameters". Below the menu bar is a tab bar with "Splash", "Report List", "Parameter Values", and "Report Viewer". The "Report Viewer" tab is active, showing a form with "Report Title" and "Category" fields, and a "Run" button at the bottom right.

THE PENDULUM EFFECT

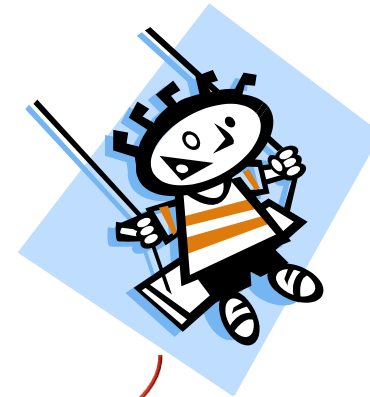
**Under-
Abstraction**



Just Right



**Over-
Abstraction**



REPORTING COMPONENT

[https://reportingServer/RunReport?id=3D2B4067-DBEE-4B3C-A5D7-DC9011300775
&user-id=24¶m1=20240531¶m2=20240606&output=pdf](https://reportingServer/RunReport?id=3D2B4067-DBEE-4B3C-A5D7-DC9011300775&user-id=24¶m1=20240531¶m2=20240606&output=pdf)

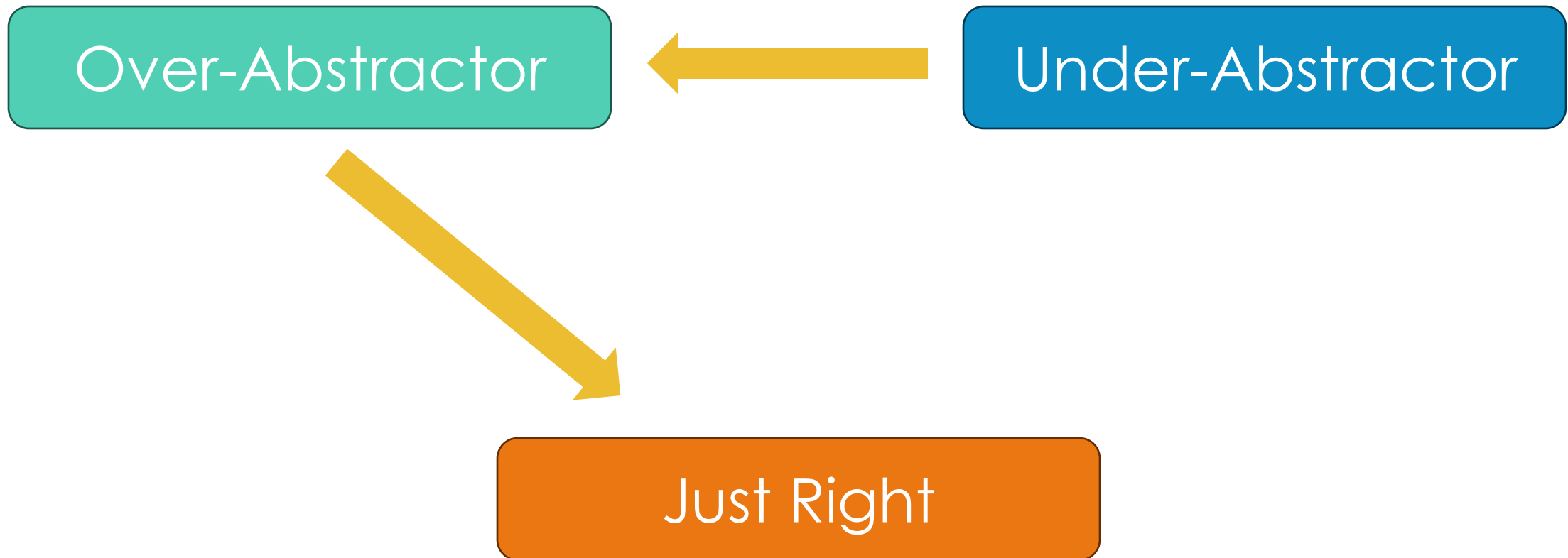
```
var report = new WebReport();  
report.Application = "Facilities";  
report.ReportName = "WeeklyMaintenance";  
report.UserName = userName;  
report.Params.Add("20240531");  
report.Params.Add("20240606");  
report.Output = ReportOutput.PDF;  
  
report.RunReport();
```

THOSE AROUND YOU

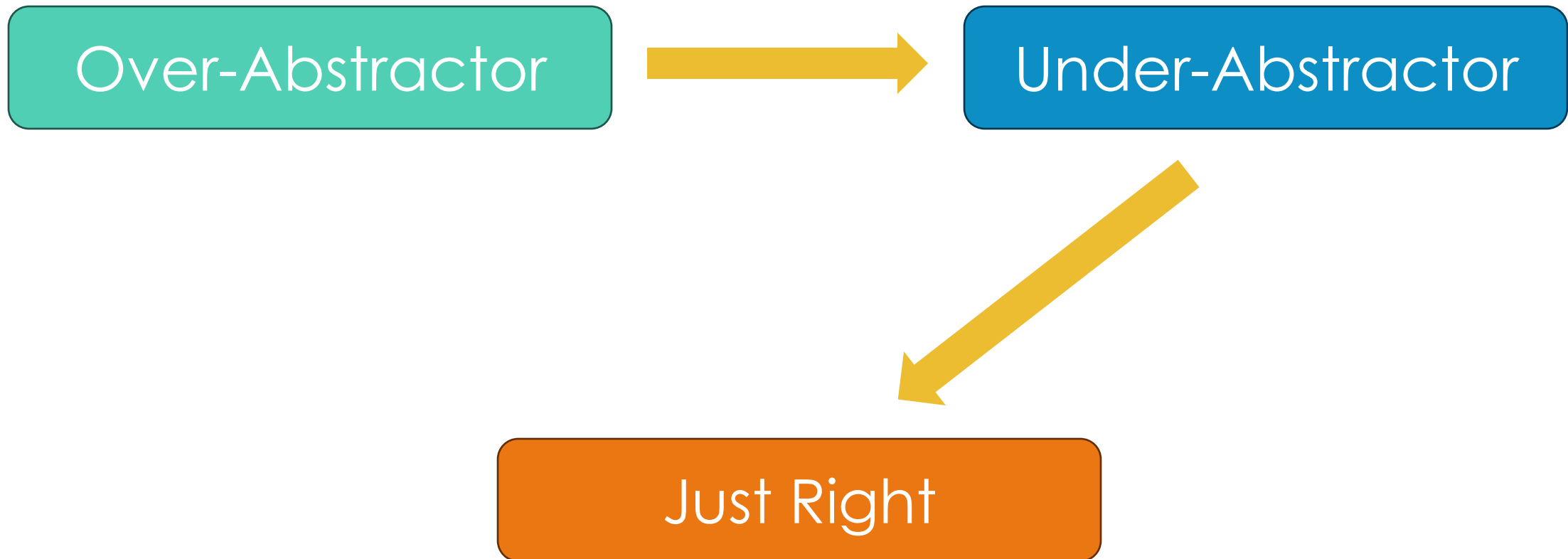
Over-Abstractor

- Jeff loved to build components.
- He liked to create code for re-use.
- He thought of all possible scenarios.

A SYMBIOTIC RELATIONSHIP



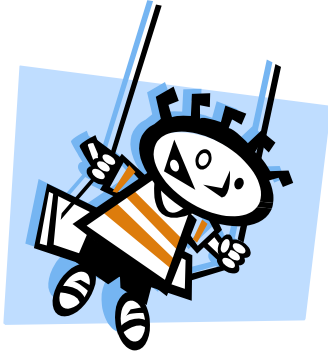
A SYMBIOTIC RELATIONSHIP



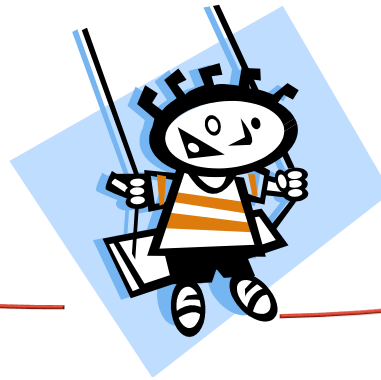


THE PENDULUM EFFECT

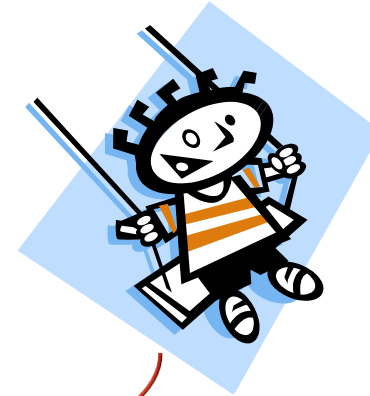
**Under-
Abstraction**



Just Right



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VARIOUS DATA SOURCES

Microsoft SQL Server

MongoDB

CSV

WebAPI

Oracle

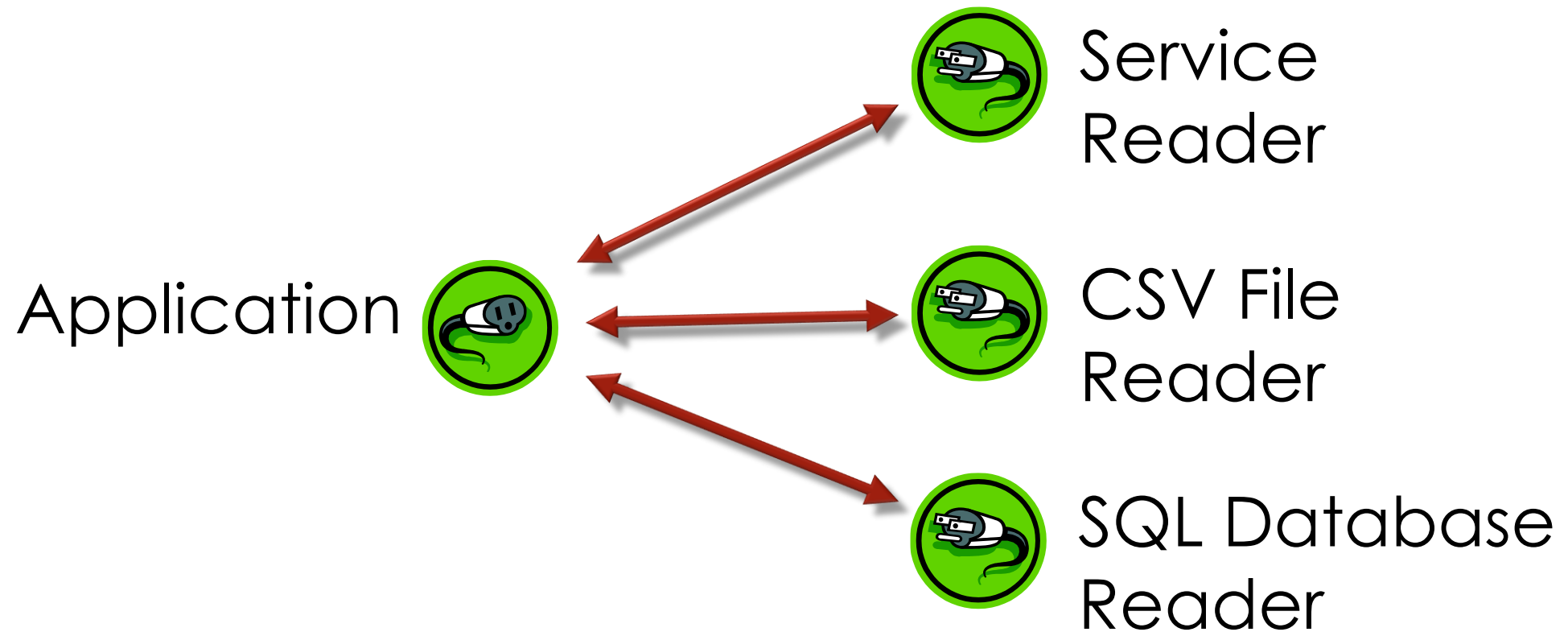
Amazon RDS

JSON

Azure Cosmos DB

Hadoop

PLUGGABLE DATA READERS





DRY

- Don't Repeat Yourself

Under-Abstractor



DON'T REPEAT YOURSELF

Consolidate
Similar
Code

Avoid
Copy/Paste

Copy/Pasta

Spaghetti
Code

SoC

- Separation of Concerns

Under-Abtractor

SINGLE RESPONSIBILITY PRINCIPLE

Complements
Separation of
Concerns

The “S” in
S.O.L.I.D.

A class should
have only one
reason to
change

A class should
do one thing
(and do it well)



YAGNI

- You Ain't Gonna Need It
- (You Aren't Going to Need It)

Over-Abstractor



MORAL OF YAGNI

- Code for the features you have now
- Add abstraction as you need it
- Don't add abstraction based on speculation

We still think about the future,
but we don't implement it yet.



KISS

- Keep It Simple, Stupid
- (Keep It Short & Simple)
- (Keep It Simple & Straightfoward)

Over-Abstractor

DDIY

- Don't Do It Yourself

Over-Abtractor

Under-Abtractor

Over-Abstractor

- Over-Abstractors like to build things to solve specific problems

Under-Abstractor

- Under-Abstractors shy away from external frameworks and libraries

EXAMPLES

Dependency Injection

- Ninject, Autofac, Spring.NET, Microsoft.Extensions.DependencyInjection

Unit Testing Framework

- NUnit, xUnit.net, MSTest, Approval Tests

Mocking

- Moq, NSubstitute, FakeItEasy, JustMock

UI Framework

- Angular, React, Vue, Prism



LEAKY ABSTRACTIONS

All non-trivial abstractions, to some degree, are leaky.

Abstractions fail. Sometimes a little, sometimes a lot. There's leakage. Things go wrong. It happens all over the place when you have abstractions.

--Joel Spolsky

The Law of Leaky Abstractions

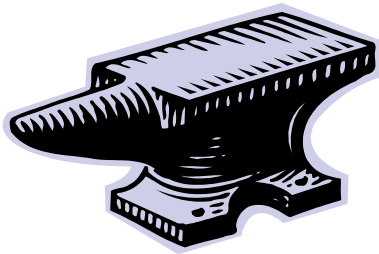
<https://www.joelonsoftware.com/2002/11/11/the-law-of-leaky-abstractions/>

ABSTRACTION IS AWESOME & AWFUL



THE GOLDBLOCKS PRINCIPLE

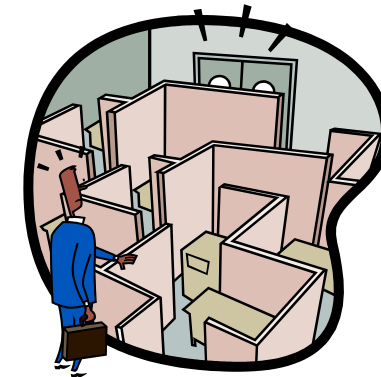
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GETTING THINGS RIGHT

DRY

- Don't Repeat Yourself

SoC

- Separation of Concerns

YAGNI

- You Ain't Gonna Need It

KISS

- Keep It Short & Simple

DDIY

- Don't Do It Yourself

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THANK YOU!

Jeremy Clark

- jeremybytes.com
- youtube.com/jeremybytes
- github.com/jeremybytes

<https://github.com/jeremybytes/vslive2025-orlando>